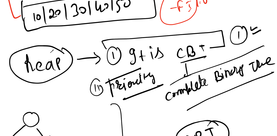
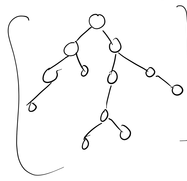
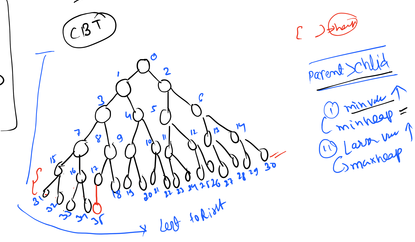
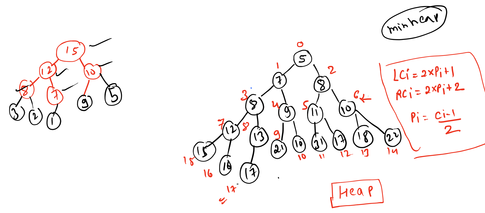
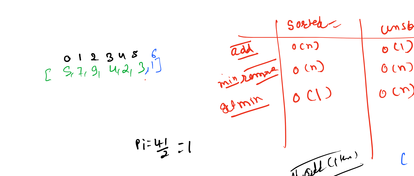
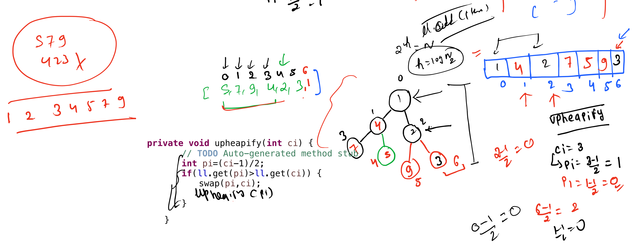




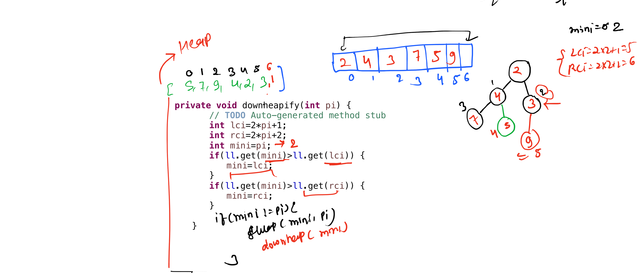




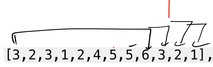




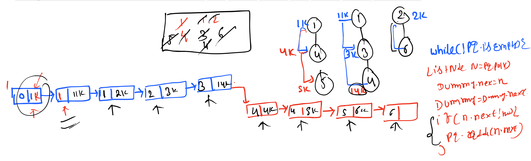

```
private void upheapify(int ci) {
    // TODO Auto-generated method stub
    int pi=(ci-1)/2;
    if(ll.get(pi)>ll.get(ci)) {
        swap(pi,ci);
        upheapify(pi);
    }
}
```

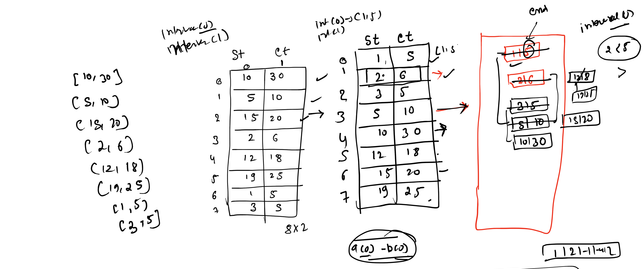


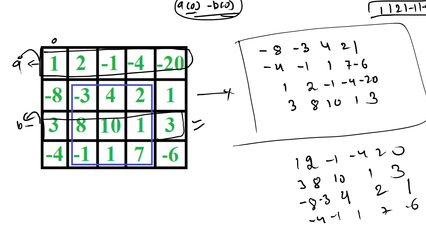

```
private void downheapify(int pi) {
        // TODO Auto-generated method stub
        int lci=2*pi+1;
        int rci=2*pi+2;
        int mini=pi; -> 2
        if(ll.get(mini)>ll.get(lci)) {
            mini=lci;
        }
        if(ll.get(mini)>ll.get(rci)) {
            mini=rci;
        }
    }  if(mini!=pi){
        swap(mini,pi)
        downheap(mini)
    }
```



Input: lists = [[1,4,5],[1,3,4],[2,6]]

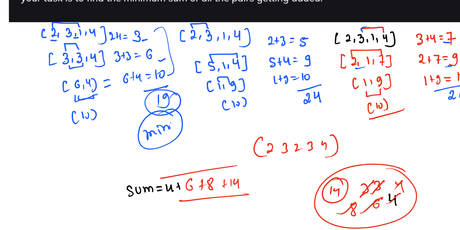






Consider we have given Array of elements a1, a2, a3, a4, at every step, we merge two-element.

let say a1 + a2 =a12

now the remaining elements would be a12, a3, a4 and we can again merge the two-element a3+a4=a34 .
now the remaining elements would be a12, a34 .

finally, we will add the last two elements to get an a1234.
your task is to find the minimum sum of all the pairs getting added.